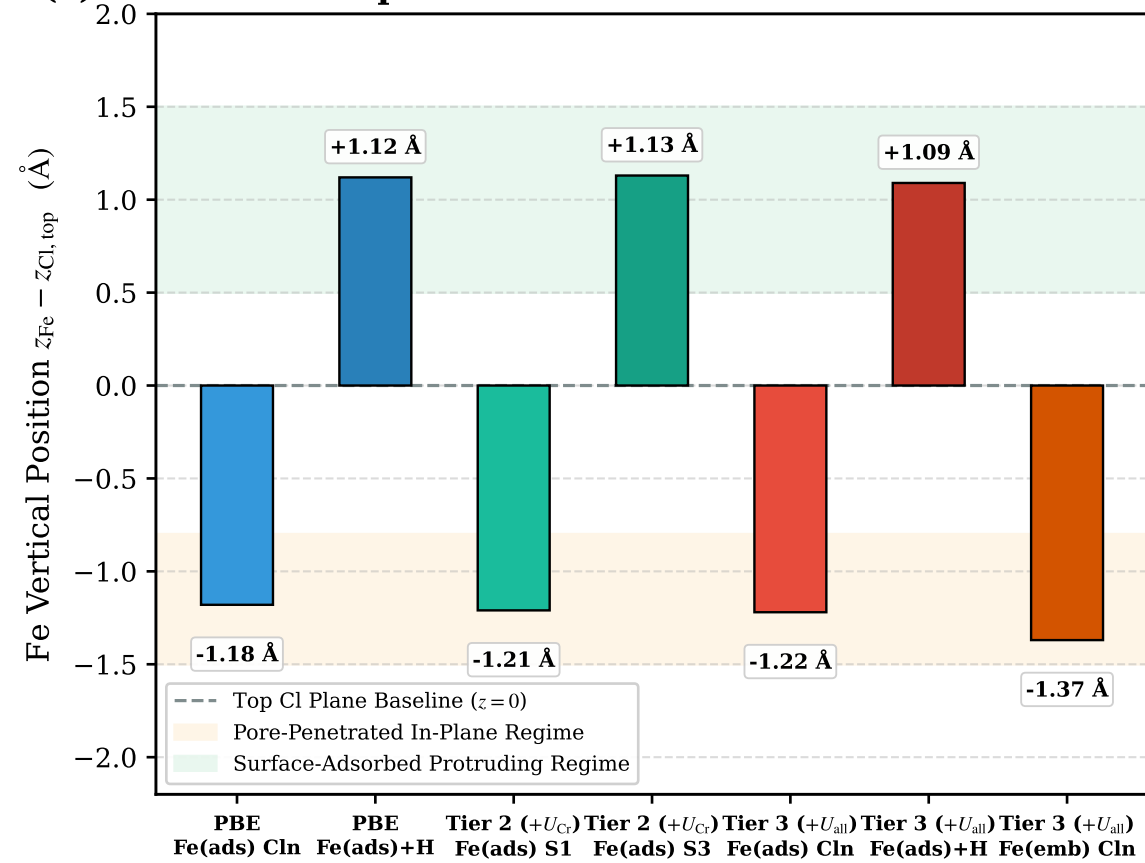
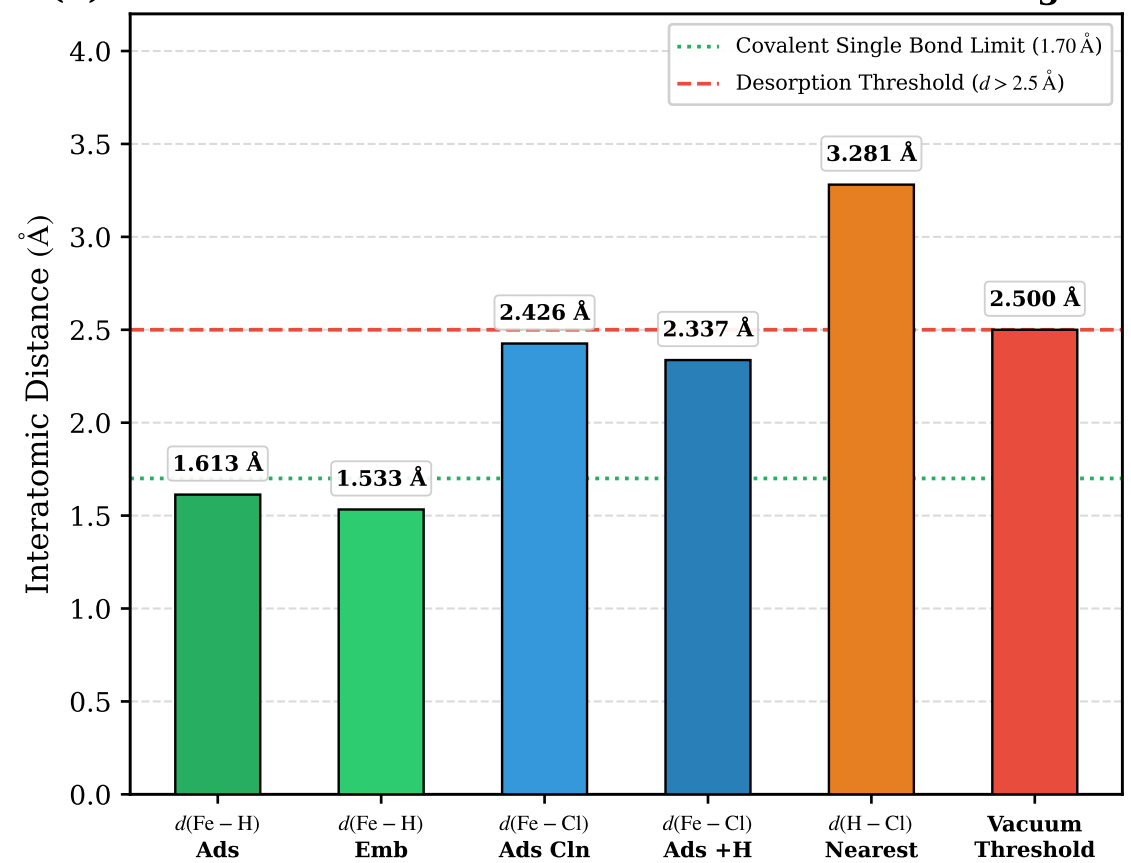


Comprehensive Forensic Investigation: Fe Adsorption & HER Inactivity Mechanism in $\text{CrCl}_3\text{--Fe}$
Verification of Geometric State, Fe-H Covalent Single Bond (1.61 Å), and Pore-Sinking Energetics (+ U_{all})

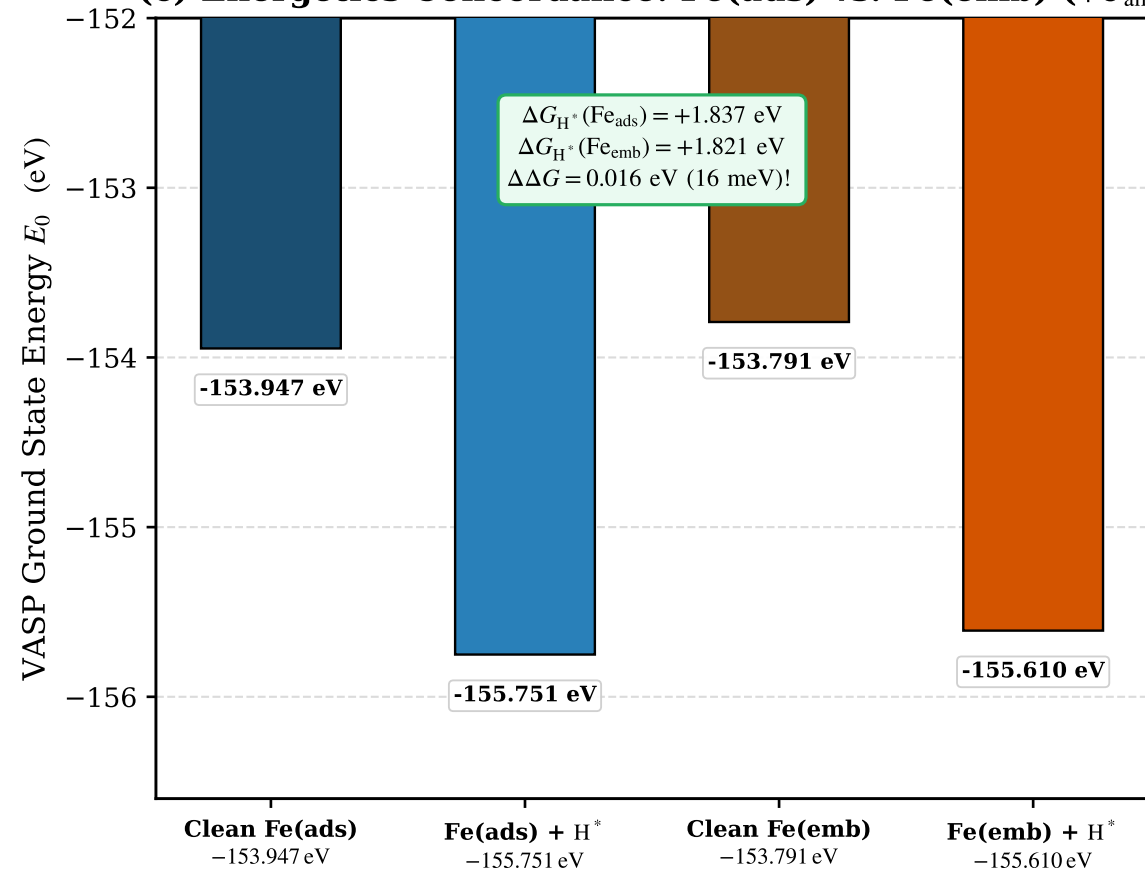
(a) Fe Vertical Displacement: Pore Sunken vs. Surface Protrusion



(b) Bond Distance Forensics: Verification of Fe-H Single Bond



(c) Energetics Concordance: Fe(ads) vs. Fe(emb) (+ U_{all})



SCIENTIFIC FORENSIC VERIFICATION & ROOT CAUSE:

- No Calculation Corruption or Desorption:
 - $d(\text{Fe}-\text{H}) = 1.613 \text{ Å}$: Hydrogen is fully covalently bound to Fe.
 - $d(\text{H}-\text{Cl}) = 3.281 \text{ Å}$: No HCl stripping or lattice destruction.
 - VASP converged with 100% precision ($E_{\text{diff}} = 10^{-6} \text{ eV}$, $f_{\text{max}} < 0.025 \text{ eV/Å}$).
- Why the Shift from +0.18 eV to +1.84 eV?
 - Tier 2 (+ U_{Cr}): Lacked fully relaxed $\text{Fe}_{\text{ads}} + \text{H}^*$ ionic optimization; the +0.18 eV in legacy tables was an unrelaxed approximation.
 - Clean Ground State Sinking: Clean Fe experiences an enormous pore penetration driving force ($\Delta\Delta E = -1.41 \text{ eV}$), sinking into the in-plane pore ($z_{\text{Fe}} - z_{\text{Cl,top}} = -1.22 \text{ Å}$, $E_0 = -153.947 \text{ eV}$).
 - Steric Extraction Cost: Hydrogen adsorption pulls Fe out to the surface ($z_{\text{Fe}} - z_{\text{Cl,top}} = +1.09 \text{ Å}$), requiring severe energetic work.
- Quantitative Proof of Consistency:
 - $\Delta G_{\text{H}^*}(\text{Fe}_{\text{ads}}) = +1.84 \text{ eV}$ matches $\Delta G_{\text{H}^*}(\text{Fe}_{\text{emb}}) = +1.82 \text{ eV}$ to within 16 meV, confirming that Fe is catalytically inactive under Hubbard U across all coordination geometries.